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The efficacy of combined physiotherapeutic scoliosis-specific exercises and manual therapy in adolescent idiopathic scoliosis

Zou Wenxia[1†], Li Yuelong[1†], Zhang Zhou[1], Jia Guoqing[1], Huang Huanjie[1], Zhang Guifang[1], Wang Chuhuai[1*], Lo Wai Leung Ambrose[1,2*] and Liu Peng[1*]

Abstract

Background Adolescent idiopathic scoliosis (AIS) is a pathological condition characterized by vertebral curvature and associated trunk deformities in adolescents. The clinical efficacy of conservative treatment in alleviating spinal curvature of AIS remains a topic of ongoing debate. The objective of this study was to investigate the impact of combined physiotherapeutic scoliosis-specific exercises (PSSE) and manual therapy (MT) on trunk deformity, spinal function, mobility, and mental health in patients with AIS.

Methods Thirty-one participants who were diagnosed with AIS whose Cobb angle was between 10–45°were enrolled in the study. Participants in the intervention group received 50 min of PSSE combined with 10 min of MT, while the control group performed 50 min of PSSE as their home exercise program. Both treatments were implemented three times a week for four weeks. Cobb angle, spinal mobility, trunk morphology (vertebral rotation angle, apical deviation, pelvic obliquity distance and angle), movement capability, and quality of life (QOL) were assessed at baseline and post intervention. The treatment effects between the intervention and control groups were analyzed using a two-way repeated measures ANOVA.

Results Following a 4-week treatment period, Cobb angle was significantly reduced from 21.58° to 18.58° in intervention group and increased from 18.00° at baseline and 19.14° post intervention in the control group. Significant improvements were also observed in spinal mobility, movement capability, quality of life, and some of the trunk morphology indices in the intervention group compared to baseline ($p < 0.05$). Improvements were significantly higher in the intervention group than the control group.

Conclusion Combining PSSE and MT shows potential benefits in alleviating AIS symptoms and improving QOL. Further studies to substantiate these findings are warranted.

Trial registration The trial was retrospectively registered in the Chinese Clinical Trial Registry (https://www.chictr.org.cn) with the registration number: ChiCTR2300071357, (Date: 12/05/2023).

Keywords Scoliosis, Manual therapy, Surface topography, Exercise therapy

†Zou Wenxia and Li Yuelong contributed equally to this work.

*Correspondence:
Wang Chuhuai
wangchuh@mail.sysu.edu.cn
Lo Wai Leung Ambrose
luowliang@mail.sysu.edu.cn
Liu Peng
liupeng2@mail.sysu.edu.cn
Full list of author information is available at the end of the article

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Background

The prevalence of adolescent idiopathic scoliosis (AIS) on a global scale has been estimated to range between 2 and 3% [1]. AIS is a structural spinal disorder that occurs during puberty and lacks a definitive etiology. Individuals who experience a high severity of lateral curvature during their peak growth period may suffer from a decreased quality of life (QOL), poor perceived appearance, disability, and pain [2, 3]. This condition not only exerts detrimental psychosocial impacts, including social barriers, elevated suicide rates in individuals with AIS [4], but also has a negative psychological effect on the parents or guardians [5]. Given the health risks associated with AIS, it is essential to identify treatment strategies that minimize the progression of the spinal curve.

To slow down the progression of AIS and correct the spinal deformity throughout pubertal growth, non-surgical treatment such as braces and physiotherapeutic scoliosis-specific exercises (PSSE) are recommended as the primary approach for mild-to-moderate AIS [6]. Bracing is considered an essential therapeutic option for patients with AIS who have Cobb angle exceeding 25°. The effectiveness of bracing may be influenced by the manufacturing techniques, brace type, adherence to wearing time and patient compliance [7, 8]. PSSE is recommended as the preferred treatment for patients with AIS who have Cobb angle of less than 25° to slow down the progression of the deformity [6]. It focuses on the integration of sensorimotor and enhancement of motor control through specific three-dimensional autocorrection and targeted exercises [9]. Several studies have suggested that PSSE leads to improvements in spinal function, health related quality of life, self-image, and psychosocial well-being [10, 11]. However, systematic reviews indicate that the evidence regarding the effectiveness of PSSE for AIS remains inconclusive [12–14].

Manual therapy (MT) can be defined as *"the use of hands to apply a force with a therapeutic intent",* which may include joint mobilization, massage, myofascial release, nerve manipulation, strain and acupressure [15]. MT of myofascial release and spinal manipulations may have the potential to reduce the Cobb angle, prevent scoliotic curve progression, improve spinal mobility, and alleviate back pain [16]. However, the effectiveness of MT in improving Cobb angle of AIS remains inconclusive. Existing evidence suggests that MT may play a role in reducing pain, decrease spinal stiffness and restore biomechanical dysfunction segments [17]. High-velocity, low-amplitude spinal manipulation (HVLA-SM) techniques have been found to elicit muscle spindle activity by evoking high-frequency peripheral sensory input [18]. Other studies also reported that spinal manipulation may facilitate cerebral cortex reorganization to enhance top-to-bottom sensorimotor integration and movement control [19]. Post-manipulation observations have shown significant increases in neuronal activity, electromyographic responses, paraspinal muscle activity, and segmental and intersegmental acceleration responses [20, 21]. The incorporation of MT may complement the benefits of exercise therapy to reduce spinal curvature of patients with AIS [22]. To date, there is a limited number of studies exploring the potential advantages of integrating MT with PSSE, and ongoing research is currently being conducted, with results awaiting publication [23]. The present study aimed to investigate the effect of combining PSSE with MT in trunk morphology, spinal mobility, movement abilities and quality of life in patients with AIS.

Materials and methods

Recruitment

Participants had been recruited through the outpatient service of The First Affiliated Hospital of Sun Yat-sen University from February 2021 to February 2022. The inclusion criteria were that study participants volunteered to be subjects of this research, that they had been diagnosed with AIS, were aged 10–18 years old at the time of the study, had a Cobb angle of 10–45° in thoracic or lumbar. Prospective participants were excluded from the study where they had non-idiopathic scoliosis, exercise contraindications, had previously received treatment for scoliosis, bracing, or could not conduct motor training. The trial was retrospectively registered at www.chictr.org.cn (ChiCTR2300071357), dated 12/05/2023. After approval from the institutional review approved by the Clinical Research Ethics Committee of The First Affiliated Hospital at Sun Yat-sen University (ETHICS No. [2021]101) dated 20/01/2021, the data collection process was started. Written informed consent was obtained from all participants and their legal representatives.

Study design and procedure

All participants were given the options to participate in either the intervention group or the control group. Standard health education was provided to both groups as part of routine clinical care. The participants in the intervention group underwent a 4-week intervention comprising 50 min of supervised PSSE and 10 min of MT sessions for three times a week. In the control group, participants and their parents were taught about the PSSE by physiotherapists, and it requires parents to record a video recording in the process in 50 min. AIS participants in the control group were asked to perform 50-min home exercise programs three times a week for four weeks, under parental supervision with video. The benefit of bracing was not investigated in the present

study since the latest Cochrane review casted uncertainty about the additional benefits of combining PSSE with bracing [24], along with supporting evidence for PSSE for mild to moderate AIS [25].

Intervention

Health education

The participants and their parents were provided with comprehensive clinical information regarding AIS. This information covered the potential adverse effects associated with AIS, the factors influencing the progression of scoliotic curvature, and the importance of seeking suitable treatment. Additionally, patients were advised to prioritize maintaining proper body posture and incorporating regular exercise into their daily routines.

Physiotherapeutic scoliosis-specific exercises

PSSE were delivered by experienced physiotherapists, which encompassed a range of techniques including spinal stabilizing exercises, breathing training, postural correction, and three dimensions autocorrection. The intervention program was based on the principles of Schroth and Scientific Exercise Approach to Scoliosis (SEAS) methods [9, 26, 27]. The levels of exercise difficulty were individualized based on specific clinical needs to achieve participants can cooperate with the exercise. Empirical evidence has shown that spinal stabilizing and core strengthening exercises effectively enhance the strength of the musculature surrounding the spine, playing a critical role in mitigating the progression of AIS [28]. During the corrective training sessions, patients were asked to maintain a neutral or corrective spinal position. Breathing training was integrated with the corrective exercises, emphasizing proper inhalation and exhalation techniques. It involved drawing air into the concave side during inhalation and expelling air from the convex side during exhalation, thus rectify the vertebral rotation and thoracic asymmetry. Daily practice of postural correction exercises enabled self-recognition and self-adjustment, potentially enhancing spinal function. Yoga balls were utilized to rectify suboptimal pelvic positions through rotational adjustments. Participants actively engaged in an exercise program involving various positions on the yoga mat and yoga balls. Figure 1 illustrates examples of PSSE exercises.

Manual therapy

Manual therapy was performed on the intervention group, which was prior to the PSSE intervention for each section. A three-minute massage was first performed to relax the myofascial membrane and muscles surrounding the spine. This initial step aimed to alleviate muscle tension and minimize the risk of injury associated with applying instantaneous pressure to the spine. It also

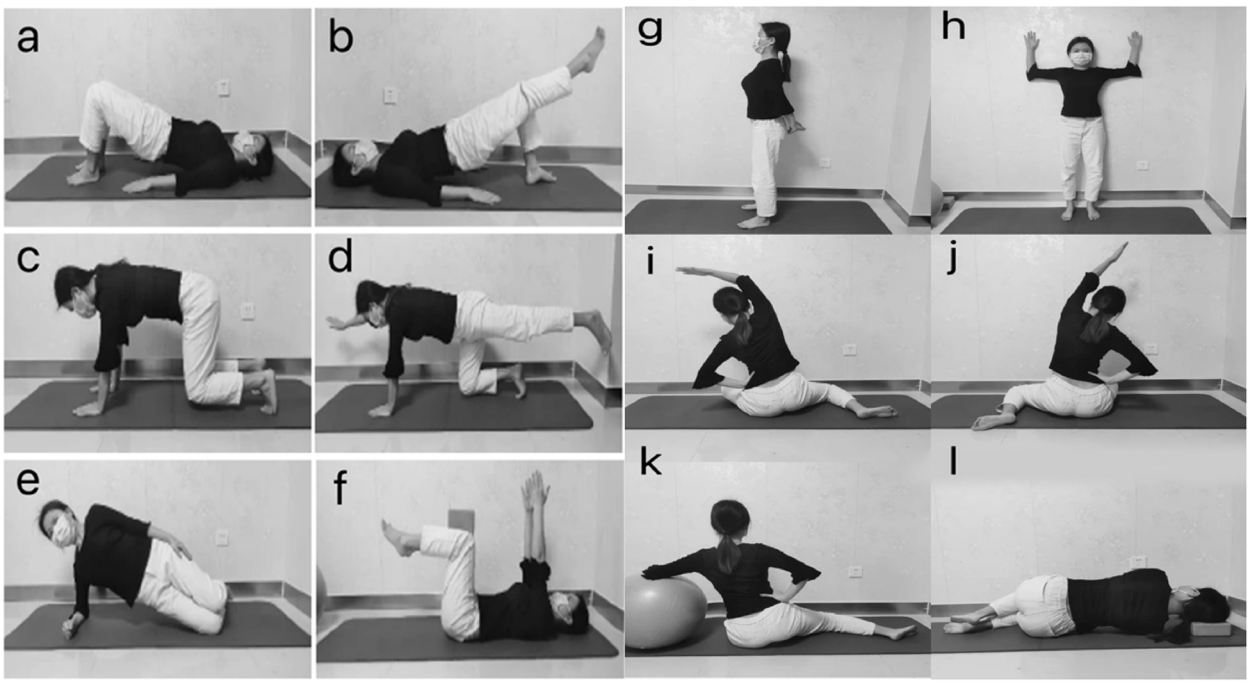


Fig. 1 **a**-**f** Illustrations of the positions for muscle strengthening exercises that primarily target core stabilization. **g**, **h** Standing with both shoulders abducted and externally rotated. **i**, **j** Intensive corrective exercises by curve stretching action movements; (**k**) breathing training (including lying and sitting)

aimed to reduce any limitations in range of movement related to soft tissue tensions. Following the massage, a high-velocity, low-amplitude thrust spinal manipulation technique was applied to the spinal level exhibiting incorrect alignment, mainly centered on the deformed top vertebra, and moving toward both ends, as determined through clinical assessment. Severe harms after HVLA-SM were relatively scarce in adolescents were reported [29].

Assessment

Outcome measures

Cobb angle was the primary outcome measure. Secondary outcome measures were spinal mobility, trunk shape, movement capability and QOL. All the outcome measures were assessed at baseline and post intervention. The assessment team consisted of three assessors who were blinded to the group allocation. Each outcome assessor was responsible for specific set of outcome measures. Each outcome measure was evaluated twice by the same assessor. The two measurements were then averaged to obtain the final value for each measure.

Cobb angle

The Cobb angle was determined by measuring the angle on the anteroposterior view of the full-spine X-ray image [30]. Cobb angle measurement was calculated from radiographs using the UniWeb Viewer (version 8.0, EBM Technologies, Taiwan, China) by an experienced physician. The end vertebrae at the top and bottom of the curvature were identified. Lines parallel to the endplates of these vertebrae were drawn and extended until they intersected. The angle at their intersection, representing the Cobb angle, was measured using the angle measurement tool and recorded to assess spinal curvature.

Spinal mobility

Spinal range of motion (ROM) was assessed using a handheld inclinometer (MicroFET3, Hoggan Scientific, Utah, USA). To measure thoracic flexion, participants were asked to stand upright with their feet shoulder-width apart. They were then asked to flex their thoracic region forward, from standing to bending, and place the inclinometer on the spinous process of C7 to take a reading of the final action. Participants were then asked to bend again, and a second reading was taken at the T12 spinous process. The difference in the readings provided the ROM of thoracic flexion. Similarly, lumbar flexion was measured from T12 to S1. Lumbar extension involved an initial reading at T12, followed by S1 after maximum extension. Lumbar lateral flexion measurements were initiated at T12, with participants leaning as far as possible to each side, followed by a second reading at S1. All of the procedures for measuring aspects of spinal mobility were performed following the prescribed guidelines to ensure consistency and standardization [31].

Trunk shape

Trunk shape was assessed using a surface topography system (DIERS Formetric4D, DIERS International GmbH, Schlangenbad, Germany) [32]. Participants stood in front of the system while maintaining a relaxed stance with their backs exposed, particularly the posterior superior iliac spine. The system automatically captured, recorded, analyzed, and generated various trunk shape indices, which show in Table 1. These measurements provided objective quantification of trunk shape. The raster-stereographic system is validated extensively against radiology, demonstrating high levels of validity and excellent intra- and inter-observer reliability in evaluating patients with AIS [33, 34].

Movement capability

Movement capability was assessed using the Functional Movement Screen (FMS), a tool designed to evaluate human movement capability in various settings without locomotion [35]. This screening tool comprises of seven movements: deep squat, hurdle step, in-line lunge, shoulder mobility test, active straight leg raises, trunk stability push-up, and rotary stability test. These movements assess flexibility, balance, coordination, and neuromuscular response abilities. A standardized testing and scoring method were employed, with scores ranging from 1 (unable to perform) to 3 (correct execution), and a score of zero was assigned if participants experienced pain while performing a subtest. The cumulative score ranges from 0 to 21, representing the total score achieved and the higher score means better status.

Quality of life

The Scoliosis Research Society-22 (SRS-22) questionnaire was employed to evaluate the QOL in participants with scoliosis [36]. The questionnaire measures QOL across five dimensions: functional activity, pain, self-image, psychological state, and treatment satisfaction. Each item is scored on a scale of one to five, total score for all dimensions is 110 and be computed, with higher scores indicating better QOL. The study utilized the Simplified Chinese version of the questionnaire [37].

Table 1 Trunk shape indexes output by the formetric 4D

Parameter	Unit	Definition[a]
Trunk length	mm	The distance from VP to the center point between DL and DR (DM)
Sagittal imbalance	mm	The distance between VP and the connecting external plumb line
Coronal imbalance	mm	The lateral distance between VP and DM
Kyphotic angle	°	The angle between the surface tangents from the ICT and ITL
Lordotic angle	°	The angle between the surface tangents from ITL and ILS
Vertebral rotation ($_{RMS}$)	°	The root mean square (RMS) of the horizontal components of the surface normal on the symmetry line
Vertebral rotation ($_{MAX}$)	°	The maximum value of the horizontal components of the surface normal on the symmetry line
Apical deviation ($_{RMS}$)	mm	The RMS deviation of the midline of the spine from the direct connection VP-DM in the frontal plane
Apical deviation ($_{MAX}$)	mm	The maximum deviation of the midline of the spine from the direct connection VP-DM in the frontal plane
Pelvic obliquity	mm	The difference in height between DL and DR
Pelvic torsion	°	The torsion of the surface normal of DL and DR

VP Vertebra Prominens, *DL* Dimple left, *DR* Dimple right, *DM* Centre between DL and DR (dimple middle), *ICT* Inflection point cervical thoracic, *ITL* Inflection point thoracic lumbar, *ILS* Inflection point lumbar sacral

[a] Trunk shape parameter definitions adapted from DIERS formetric III 4D Manual (Created 21.06.2010, Revision grade 5)

Statistics

Statistical analysis was conducted in SPSS v.20 (IBM SPSS, Armonk, NY, USA). The distribution of the data was assessed using the Kolmogorov–Smirnov test. Two-way analysis of variance (ANOVA) was conducted to assess the effects of groups and different time points. In cases where the main effect of time was significant, post hoc tests with Bonferroni were performed. Additionally, the change rate (obtained by dividing the difference value by the baseline value) and change value of each indicator in the two groups before and after treatment were calculated and analyzed using independent samples t-test. The minimal clinically important difference (MCID) for Cobb angle and SRS-22 for spinal deformity patients with non-surgical treatments are 5° and 0.11 respectively [38, 39]. For the FMS, spinal range of motion, trunk shape, to detect if changes are clinically meaningful, the MCID was obtained by multiplying the standard deviation of the baseline scores by 0.2 (i.e., MCID = 0.2 × SD) [40]. The statistical significance level was set at $P = 0.05$.

Results

A total of 31 participants with AIS were recruited. Seventeen participants enrolled in the intervention group and fourteen participants enrolled in the control group. There are twelve participants with Risser 0–2 and five participants with over Risser 3 in the intervention group, while ten participants with Risser 0–2 and four participants with over Risser 3 in the control group. The curve types of participants in the intervention group included nine Thoracic curvature, two Lumbar curvature, two Thoracolumbar curvature and four Double major curvature, while three Thoracic curvature, four Lumbar curvature, six Thoracolumbar curvature and one Double major curvature in the control group. No significant difference was observed between the two groups for gender, age, height, weight, and Cobb angle ($p < 0.05$) in Table 2.

For Cobb angle, the results of two-way ANOVA analyses indicated the main effect of group, and the main effect of time was not significant ($p > 0.05$). A significant interaction effect was observed between the two groups at different time points ($p < 0.05$). For FMS and SRS-22, the main effect of the group was not significant, while the interaction effects between the two groups at different time points were significant ($p < 0.05$). Table 3 presents the results of ANOVA analysis. The t-test results for change rate (difference value/baseline value) and change value of Cobb angle were significantly different between the two groups ($p < 0.05$). The change rate and value of FMS and SRS-22 were also significantly different

Table 2 Characteristics of the sample cohorts (mean (SD))

Demographics	Interventions ($n = 17$)	Controls ($n = 14$)	*P*-value
Gender (M: F)	6:11	2:12	$p = 0.240$
Age (years)	12.53(1.32)	13.43(2.27)	$p = 0.119$
Height (cm)	162.35(10.53)	157.86(8.90)	$p = 0.222$
Weight (kg)	44.76(7.74)	46.43(6.64)	$p = 0.495$
Cobb angle (°)	21.58(8.15)	18.00(5.42)	$p = 0.170$

SD Standard deviation, *P*-value: probability value, Statistically significant level was set at $P = 0.05$

between the two groups ($p < 0.05$). Table 4 presents the t-test results of change rate/value and MCID. The Cobb angle of the intervention group and the control group did not reach the MCID threshold. For FMS and SRS-22 total scores, the intervention group reached the MCID threshold and controls group did not.

The main between group effect for thoracic spine and lumbar spine ROM were not significantly (Table 5). Significant differences were observed in lumbar flexion and lumbar left lateral flexion at different time points in the intervention group. The t-test results indicated significant differences ($p < 0.05$) in the change rate and value of spinal ROM between the two groups (Table 6). The ROM of spinal including thoracic flexion, lumbar flexion, lumbar extension, lumbar left and right lateral flexion of intervention group reached the MCID threshold, while control group did not. Significant differences were observed in apical deviation ($_{RMS}$)(mm) and pelvic obliquity(mm) at different time points in the intervention group (Table 7). The results of t-test for the change rate and change values of some trunk shape indices (Table 8) between the two groups were significant ($p < 0.05$). Trunk shape indices of surface topography including vertebral rotation (RMS)(°), vertebral rotation(MAX)(°), apical deviation (RMS)(mm), apical deviation(MAX)(mm), sagittal imbalance(mm), coronal imbalance (mm), kyphotic angle(°), lordotic angle(°), pelvic torsion(°) and pelvic obliquity(mm) of intervention group reached the MCID threshold, while the control group did not.

Table 3 The results of ANOVA for Cobb angle, FMS score, and SRS-22 total scores at baseline and post-intervention

Assessment	Interventions ($n = 17$) mean (SD)		Controls ($n = 14$) mean (SD)		Main effect of time		Main effect of group		Interaction effects	
	Primary	4 Weeks	Primary	4 Weeks	F time	P time	F group	P group	$F_{Group*Time}$	$P_{Group*Time}$
Cobb angle (°)	21.58(8.15)	18.58(9.68)	18.00(5.42)	19.14(6.43)	2.651	0.114	0.307	0.584	13.190	0.001
FMS (score)	13.52(2.23)	15.11(2.14)	14.78(1.25)	14.78(1.25)	9.664	0.004	0.574	0.455	9.664	0.004
SRS-22 (score)	85.82(8.16)	89.82(7.55)	87.42(7.49)	86.57(6.99)	5.290	0.029	0.096	0.759	12.635	0.001

FMS Functional Movement Screen, *SRS-22* Scoliosis Research Society-22, *SD* Standard deviation, *F* F ratio, *P* probability value

Table 4 The results of t-test for the change rate and value, MCIDs of Cobb angle, FMS score, and SRS-22 total scores

Assessment	MCID	The change value				The change rate			
		Interventions ($n = 17$) mean (SD)	Controls ($n = 14$)	*t*	*p*	Interventions ($n = 17$) mean (SD)	Controls ($n = 14$)	*t*	*p*
Cobb angle (°)	5.00	3.00(3.16)	-1.14(3.15)	3.632	0.001	0.16(0.20)	-0.06(0.17)	3.435	0.002
FMS (score)	0.44	1.58(1.83)	0.00(0.55)	3.379	0.003	0.13(0.15)	0.001(0.03)	3.302	0.004
SRS-22 (score)	2.42	4.00(4.41)	-0.85(2.82)	3.555	0.001	0.04(0.05)	0.008(0.03)	3.455	0.002

FMS Functional Movement Screen, *SRS-22* Scoliosis Research Society-22, *MCID* Minimal clinically important difference, *SD* Standard deviation, *t* t-value, *p* probability value

Table 5 The results of ANOVA for ROM of spinal at baseline and post-intervention

Assessment	Interventions ($n = 17$) mean (SD)		Controls ($n = 14$) mean (SD)		Main effect of time		Main effect of group		Interaction effects	
	Primary	4 Weeks	Primary	4 Weeks	F time	P time	F group	P group	$F_{Group*Time}$	$P_{Group*Time}$
Thoracic flexion (°)	19.47(10.00)	27.88(5.82)	24.35(11.73)	22.42(8.10)	4.724	0.220	0.009	0.923	12.017	0.002
Lumbar flexion (°)	66.58(10.45)	79.88(15.88)	68.57(11.79)	66.85(12.64)	12.800	0.001	1.595	0.217	21.502	< 0.001
Lumbar extension (°)	18.94(8.08)	25.17(5.49)	25.71(5.39)	23.42(3.22)	1.976	0.170	2.418	0.131	9.196	0.005
Lumbar left lateral flexion (°)	19.70(7.05)	32.85(5.25)	27.71(6.30)	25.28(4.93)	22.852	< 0.001	0.016	0.899	48.338	< 0.001
Lumbar right lateral flexion (°)	21.17(9.19)	29.88(4.97)	25.85(7.95)	22.78(4.80)	6.133	0.019	0.283	0.599	26.797	< 0.001

ROM Range of motion, *SD* Standard deviation, *F* F ratio, *P* Probability value

Table 6 The results of t-test for the change rate and value, MCIDs of ROM of spinal

Assessment	MCID	The change value				The change rate			
		Interventions ($n=17$)	Controls ($n=14$)	t	p	Interventions ($n=17$)	Controls ($n=14$)	t	p
		mean (SD)				mean (SD)			
Thoracic flexion (°)	2.00	8.41(10.14)	-1.92(5.06)	3.467	0.002	0.86(1.11)	-0.01(0.19)	3.188	0.005
Lumbar flexion (°)	2.09	13.29(11.46)	-1.71(4.19)	5.005	$P<0.001$	0.20(0.19)	-0.02(0.06)	4.728	$P<0.001$
Lumbar extension (°)	1.61	6.23(9.64)	-2.28(4.54)	3.232	0.004	0.57(0.67)	-0.05(0.23)	3.610	0.002
Lumbar left lateral flexion (°)	1.41	13.11(7.08)	-2.42(4.87)	6.953	$P<0.001$	0.94(0.93)	0.06(0.14)	4.396	$P<0.001$
Lumbar right lateral flexion (°)	1.81	8.70(6.95)	-3.07(5.39)	5.177	$P<0.001$	0.84(1.37)	0.07(0.18)	U = 21.00	$P<0.001$

ROM Range of motion, *MCID* Minimal clinically important difference, *SD* Standard deviation, *t* t-value, *p* probability value

Discussion

The present study extends the current body of knowledge by evaluating the effects of integrating MT and PSSE in patients with AIS. It incorporates a range of outcome measures to investigate trunk morphology, spinal mobility, movement abilities, and quality of life. Improvements in Cobb angle, QOL, movement capability, spinal mobility and some of the trunk shape indices were observed in participants who received 4 weeks of combined PSSE and MT interventions. The change value and change rate of the recorded parameters were also larger in the intervention group than the control group, demonstrating the potential benefits associated with combining PSSE and MT in managing AIS.

Combined MT and PSSE on scoliosis curvature

A significant change in Cobb angle was observed in the intervention group (mean Cobb: 21.58° ± 8.15° before intervention; 18.58° ± 9.68° after intervention). Although the absolute change in Cobb angle did not exceed the MCID in the intervention group, the change was consistent with the amount of change induced by PSSE (mean decrease of 3.04 degrees) reported in a recent Cochrane review [24]. The changes in Cobb angle from the PSSE and MT group was also higher than those of home exercise programs alone, which demonstrated a deterioration in both Cobb angle and trunk shape indices. This finding indicated the potential benefit of combining PSSE and MT in slowing or halting the progression of scoliosis compared to home exercise alone. Spinal mobility, including thoracic and lumbar flexion, lumbar lateral flexion, and posterior extension, along with higher FMS scores, exceeded the MCID in the intervention group but not in the control group. Trunk shape indices including spinal rotation angle, distance from the median line of vertebral migration, and pelvic tilt, also exceeded the calculated MCID. These findings suggest that supervised PSSE may improve surface topography characteristics in patients with AIS in the short term [41]. The pathogenesis of AIS is believed to involve characteristics of muscle asymmetry and muscle tension differences between the left and right sides of the spine, resulting in spinal imbalance in the sagittal and frontal planes [42, 43]. MT was shown to adjust tension along the myofascial chain of the spine, promoting overall balance recovery [44], and may improve spinal range of motion by adjusting facet joint positions [17, 45]. Additionally, decreased muscle spindle activities in AIS patients [46] can be stimulated by MT [18, 47]. Reduced muscle strength in AIS is linked to higher intervertebral disc pressure, contributing to scoliosis [48]. Strengthening core muscles through PSSE may reduce intervertebral disc pressure [49] and adjust spinal tension integrity, decreasing scoliosis risk [26, 27]. Postural balance deficits contribute to AIS development [50], and PSSE can correct these deficits through exteroceptive and proprioceptive stimulation [51, 52].

PSSE as home exercise on scoliosis curvature

The control group received PSSE program as home exercises without physiotherapist supervision. The findings from the present study did not demonstrate significant benefits in reducing or slowing the progress of spinal curvature in patients with AIS. PSSE is recommended as the first line of treatment for scoliosis due to supporting evidence of its effectiveness. However, conflicting results have been reported in several studies. A study by Negrini et al. showed that Cobb Angle increased by 1.70° ± 7.24° during 25 months of treatment ($p<0.05$ between groups) [53]. In a prospective study, Zapata and co-authors noted that the difference in Cobb angle between the PSSE group and observation group was not significant after six months [54]. These findings were consistent with the

Table 7 The results of ANOVA for trunk shape index of Surface topography at baseline and post-intervention

Assessment	Interventions ($n=17$) mean (SD)		Controls ($n=14$) mean (SD)		Main effect of time		Main effect of group		Interaction effects	
	Primary	4 Weeks	Primary	4 Weeks	F time	P time	F group	P group	$F_{Group*Time}$	$P_{Group*Time}$
Trunk length(mm)	406.10(45.44)	412.04(44.21)	404.22(39.05)	401.29(33.49)	0.880	0.356	0.182	0.673	7.598	0.010
Vertebral rotation ($_{RMS}$)(°)	5.95(2.57)	4.60(2.08)	5.46(1.84)	5.76(1.65)	2.439	0.129	0.245	0.624	5.975	0.021
Vertebral rotation ($_{MAX}$)(°)	11.21(4.58)	8.37(3.14)	9.95(3.18)	10.78(2.80)	3.169	0.086	0.251	0.620	10.635	0.003
Apical deviation ($_{RMS}$)(mm)	8.26(4.86)	4.99(2.41)	8.03(4.30)	7.53(4.54)	12.385	0.001	0.693	0.412	6.669	0.015
Apical deviation ($_{MAX}$)(mm)	15.06(8.56)	8.90(3.68)	14.61(7.23)	12.44(7.54)	17.506	<0.001	0.446	0.509	4.026	0.054
Sagittal imbalance (mm)	2.77(1.58)	2.24(1.80)	2.95(1.61)	3.00(1.29)	1.169	0.288	0.779	0.385	1.637	0.211
Coronal imbalance (mm)	7.62(4.62)	5.16(3.25)	8.13(5.48)	8.39(6.94)	0.913	0.347	1.671	0.206	1.381	0.249
Kyphotic angle (°)	36.29 (10.36)	33.27 (8.85)	37.70 (8.78)	36.44 (11.19)	3.918	0.057	0.459	0.503	0.666	0.421
Lordotic angle (°)	40.97 (8.38)	38.57 (10.25)	34.43 (10.68)	34.70 (10.25)	0.551	0.464	2.548	0.121	0.861	0.361
Pelvic torsion (°)	2.66(1.29)	2.34(1.66)	1.92(1.17)	2.14(1.07)	0.049	0.826	1.205	0.281	1.451	0.238
Pelvic obliquity(mm)	2.32(2.46)	0.96(1.26)	2.19(1.73)	2.20(1.91)	3.781	0.062	0.871	0.358	3.861	0.059

SD Standard deviation, *F* F ratio, *P* Probability value

Table 8 The results of t-test for the change rate and value, MCIDs of trunk shape index of Surface topography

Assessment	MCID	The change value				The change rate			
		Interventions ($n=17$)	Controls ($n=14$)	t	p	Interventions ($n=17$)	Controls ($n=14$)	t	p
		mean (SD)				mean (SD)			
Trunk length(mm)	9.08	5.94(8.05)	-2.92(9.87)	2.757	0.010	0.15(0.01)	-0.006(0.02)	2.778	0.009
Vertebral rotation ($_{RMS}$)(°)	0.51	1.35(1.82)	-0.29(1.92)	2.444	0.021	0.16(0.25)	-0.14(0.47)	2.334	0.027
Vertebral rotation ($_{MAX}$)(°)	0.91	2.84(3.03)	-0.83(3.24)	3.261	0.003	0.21(0.19)	-0.15(0.39)	3.186	0.005
Apical deviation ($_{RMS}$)(mm)	0.97	3.27(3.71)	0.50(1.63)	2.582	0.015	0.33(0.19)	0.05(0.25)	3.560	0.001
Apical deviation ($_{MAX}$)(mm)	1.71	6.16(6.73)	2.16(3.46)	2.007	0.054	0.34(0.20)	0.13(0.28)	2.386	0.024
Sagittal imbalance (mm)	0.31	0.53(1.00)	-0.04(1.51)	1.280	0.070	0.17(0.45)	-0.44(1.26)	1.743	0.101
Coronal imbalance (mm)	0.92	2.45(6.33)	-0.25(6.46)	1.175	0.249	-1.12(4.18)	-0.53(1.56)	-0.494	0.625
Kyphotic angle (°)	2.07	3.02(5.68)	1.25(6.34)	0.816	0.421	0.06(0.15)	0.03(0.17)	0.518	0.608
Lordotic angle (°)	1.67	2.38(8.62)	-0.26(6.89)	0.928	0.361	0.04(0.20)	-0.04(0.32)	0.976	0.337
Pelvic torsion (°)	0.25	0.31(1.23)	-0.22(1.24)	1.205	0.238	0.07(0.37)	-1.54(5.25)	1.155	0.269
Pelvic obliquity(mm)	0.49	1.35(2.37)	0.007(1.16)	2.089	0.047	0.67(0.61)	0.08(0.60)	2.940	0.008

MCID Minimal clinically important difference, *SD* Standard deviation, *t* t-value, *p* probability value

findings of the present study where no significant difference in scoliosis curvature was observed post intervention. The negative results may be attributed to the difference in exercise delivery settings. A previously published randomized controlled trial reported that PSSE program, when applied in the clinic under physiotherapist supervision, was superior to the home exercise [54]. The study by Kuru et al., (2015) reported an increase in Cobb angle of 5 degrees and angle of trunk rotation of 1.5 degrees for the home exercise group [55]. Thus, the capacity for each child to learn and perform exercises effectively in home setting, potentially coupled with poor compliance, may contribute to the negative results of PSSE [56, 57].

PSSE has the characteristics of strong feasibility, but the specific programs are varied. Many studies have explored the efficacy of PSSE combined with other treatment programs in the treatment of AIS, and the results are mostly positive. Schreiber et al. found that PSSE, when used in combination with conventional therapy, improved the Cobb angle, vertebral rotation, and increased back muscle endurance and body posture in AIS [58]. Rrecaj-Malaj et al. showed that PSSE and Pilates exercises provided benefit on the Cobb angle, angle of trunk rotation (ATR), chest expansion, trunk flexion and QOL in mild and moderate AIS [59].

Quality of life and cosmetic appearance

Spinal curve deformities in patients with AIS can lead to both physical and psychological complications [60]. The International Society on Scoliosis Orthopedic and Rehabilitation Treatment published a consensus report emphasizing that the primary goals in treating AIS should be deformity correction and aesthetic appearance enhancement [6]. Gera et al., found that using a combination of spine adjustment techniques and PSSE can significantly reduce the psychosomatic symptoms in functional scoliosis [61]. This finding aligns with the results of our study, where the combination of PSSE and MT may have better outcomes than home-based PSSE alone. The intervention group demonstrated a 4-point increase in quality-of-life total scores and exceeded the MCID value, whereas the home-training group demonstrated a 1-point decrease. This finding is consistent with a study where home exercise alone was found to have no significant effect on QOL [62]. Our findings also shed light on cosmetic and physical improvements in the participants who received combined MT and PSSE. Participants in the intervention group exhibited more significant enhancements than participants in the control group in the spinal appearance, potentially due to the increased symmetry and reduction in spinal curve. This enhancement in appearance corroborates with the observed improvement in quality of life, further supporting the efficacy of combined intervention of MT and PSSE.

Limitations

The results of the present study should be interpreted with cautions due to the limitations. The absence of a control group, comprising either a general exercise group or individual PSSE and MT training groups, limits the direct comparisons which hinder the interpretation on the specific impact of each intervention. However, follow-up data and imaging data lend

quantitative support to our positive results. Compliance to PSSE was not monitored in the present study. This may lead to incomplete or insufficient treatment in both groups. Factors such as high academic pressure or inadequate parental supervision are known to influence compliance to the prescribed regimen. Center-specific factors and the limited representation of patients with AIS within the sample may impede the external validity of the findings. The sample size of 31 participants may be considered small, which affects the generalizability of the findings. Given the non-randomized grouping in this study, the influence of the families' choices on the experimental outcomes should be considered. The four-week intervention period limits the interpretation of the potential long-term benefits. Future studies should incorporate longer follow-up periods to evaluate the long-term benefits of the intervention regimen. The inclusion of participants with potentially non-progressive curves and curvature of less than 20° may mask or exaggerate the observed benefit. Observational approach may often be adopted for patients with mild curvature due to potential of non-progression. However, this approach is contentious and lacks definitive guidance for untreated curves under 20° due to a lack of clinical indicators for scoliosis progression [63]. The observation approach does not address the patient's anxiety and the stress of the parents, who worry about the progression of curvatures [64]. Thus, early identification and early intervention may be beneficial. The SOSORT guideline which indicated that current evidence suggested that conservative treatment was effective at stopping curve from progression, as well as improving the curves at skeletal maturity [6]. Stratified sampling was not conducted, and Risser stage and Cobb angle size were not applied as influencing factors to assess the intervention's efficacy for different severities of scoliosis. Future studies should incorporate stratified sampling and consider Risser stage and Cobb angle size to rigorously evaluate the efficacy across varying severities of scoliosis.

Conclusion

The present study provides evidence supporting the potential clinical benefits of integrating PSSE with MT for managing AIS. The combined intervention approach demonstrated improvements in Cobb angle, spinal mobility, movement ability, trunk morphology, and quality of life. However, these findings should be interpreted with caution due to study limitations. Larger clinical trials are recommended to substantiate the findings of this study and to further explore the long-term benefits of the combined approach.

Abbreviations

AIS	Adolescent idiopathic scoliosis
PSSE	Physiotherapeutic scoliosis-specific exercises
MT	Manual therapy
QOL	Quality of life
ROM	Range of motion
HVLA-SM	High-velocity, low-amplitude spinal manipulation
SEAS	Scientific Exercise Approach to Scoliosis
FMS	Functional Movement Screen
ANOVA	Two-way analysis of variance
SRS-22	Scoliosis Research Society-22
ATR	Angle of trunk rotation
MCID	Minimal clinically important difference

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Authors' contributions

ZWX, LYL, WCH, LWLA, and LP: Study conception and design ZZ, JGQ, HHJ, and ZGF: Acquisition of data and figures ZWX, LP, and LYL: Analysis and interpretation of data ZWX: Drafting of the manuscript WCH, LWLA, and LP: Critical revision and proofreading of the manuscript All authors: final approval of the manuscript.

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Data availability

Data that support the findings of this study have been deposited in the Scientific Research System of Affiliated Hospital of Sun Yat-sen University, China.It is available from the authors upon request.

Declarations

Ethics approval and consent to participate

The trial was retrospectively registered at www.chictr.org.cn (ChiCTR2300071357), dated 12/05/2023. Approval from the institutional review approved by the Clinical Research Ethics Committee of The First Affiliated Hospital at Sun Yat-sen University (ETHICS No. [2021]101) dated 20/01/2021. Written informed consent was obtained from all the participants.

Consent for publication

Written informed consent for patient information and images to be published was provided by the participants.

Competing interests

The authors declare no competing interests.

Author details

[1]Department of Rehabilitation Medicine, The First Affiliated Hospital, Sun Yat-Sen University, No. 58 Zhongshan Er Road, Guangzhou 510080, P.R. China. [2]Guangdong Engineering and Technology Research Centre for Rehabilitation Medicine and Translation, The First Affiliated Hospital, Guangzhou, China.

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